

Research Interests

Human-Computer Interaction, Human-AI Interaction, Visualization, Accessibility

Education

MS/PhD University of Massachusetts Amherst, Computer Science

Sep 2024 - Present

BS Sungkyunkwan University, Computer Science and Engineering

Mar 2019 - Feb 2024

The University of Texas at Austin (Exchange Student Program)

Jan - May 2022

Publications

Peer-Reviewed Conference and Journal

1. RubRIX: Rubric-Driven Risk Mitigation in Caregiver-AI Interactions [🔗](#)

- Drishti Goel, **Jeongah Lee**, Qiuyue Joy Zhong, Violeta J. Rodriguez, Daniel S. Brown, Ravi Karkar, Dong Whi Yoo, Koustuv Saha
- 2026 Findings of the Association for Computational Linguistics (ACL)

2. From Text to Visuals: Using LLMs to Generate Math Diagrams with Vector Graphics [🔗](#)

- Jaewook Lee, **Jeongah Lee**, Wanyong Feng, Andrew Lan
- AIED 2025: 26th International Conference on Artificial Intelligence in Education

3. IoT Edge-Cloud: An Internet-of-Things Edge-Empowered Cloud System for Device Management in Smart Spaces [🔗](#)

- Yoseop Joseph Ahn, Minje Kim, **Jeongah Lee**, Yiwen Shen, Jaehoon Paul Jeong
- 2023 IEEE Network Magazine

Preprints and Under Review

1. Resolving Ruptures: Mental Health Professionals' Perspectives on How LLMs Identify and Resolve Conversational Ruptures

- **Jeongah Lee**, Qiuyue Joy Zhong, Drishti Goel, Violeta J. Rodriguez, Dong Whi Yoo, Koustuv Saha, Ravi Karkar

2. Understanding How Gratitude Practices Evolve in Everyday Life

- Qiuyue Joy Zhong, **Jeongah Lee**, Drishti Goel, Violeta J. Rodriguez, Dong Whi Yoo, Koustuv Saha, Ravi Karkar

2. Inform, Coach, Relate, Listen: Auditing LLM Caregiving Support Responses [🔗](#)

- Drishti Goel, Agam Goyal, Veda Duddu, Olivia Pal, **Jeongah Lee**, Qiuyue Joy Zhong, Violeta J. Rodriguez, Daniel S. Brown, Dong Whi Yoo, Ravi Karkar, Koustuv Saha

3. En3: A Tripartite Approach Toward Holistic Understanding and Detection of Engagement in Educational Video Content

- Sidharth Anupkrishnan, Itir Sayar, **Jeongah Lee**, Sri Harsha Musunuri, Guan-Ming Su, Tam Vu, Madeline Endres, Ravi Karkar, Phuc Nguyen

4. Conversational AI for Floorplan Accessibility: How Blind and Low-Vision Users Explore Space Through Dialogue

- **Jeongah Lee**, Sriikiran Kavuri, Hari Iyer, Ali Sarvghad

5. Cohort-based Semantic Labeling: AI-Enabled Recovery of Visualization Semantics from Deployed SVGs [🔗](#)

- **Jeongah Lee**, Hima Varshini Surisetty, Durga Nirmaleswaran, Jahnvi Sharma, Sriikiran Kavuri, Narges Mahyar, Ali Sarvghad

Research & Work Experiences

Dolby Advanced Technology Group, PhD Research Intern

Atlanta, GA, USA

- Advisor: Dr. Gautam Bhattacharya, Dr. Feiyu Lu, Dr. Jian Chen
- Conducted research on designing proactive AI agents for spatial audio mixing

HCI-VIS lab, WHAT lab @ University of Massachusetts Amherst, Graduate Research Assistant

Amherst, MA, USA

Sep 2024 - Present

- Advisor: Dr. Ravi Karkar [🔗](#), Dr. Ali Sarvghad [🔗](#)
- Led research on AI-driven mental health tools for Alzheimer's caregivers, integrating journaling, conversational interaction, and cognitive reframing (CBT) support

Seoul National University Bundang Hospital, Natural Language Processing (NLP) Researcher

Bundang, Korea

Mar - Jul 2024

- Developed a predictive model for lung cancer TNM staging using Electronic Health Record (EHR) datasets, improving Macro F1-score from 0.70 to 0.93 through prompt design and data augmentation

Cipherome, Inc., Machine Learning (ML) Engineer Intern

San Jose, CA, USA

Mar - Aug 2023

- Developed the pipeline for an ML module within a clinician-focused medical data analysis platform
- Designed Figma wireframes to improve the platform's user interface and experience (UI/UX)

IoT lab @ Sungkyunkwan University, Undergraduate Student Researcher

Suwon, Korea

Jul - Nov 2021

- Advisor: Dr. Jaehoon Paul Jeong [🔗](#)
- Developed an IoT edge-empowered Cloud System for the visual control of IoT devices in a user's smart-

phone, improving location tracking accuracy by 39% through the integration of Smart Pedestrian Dead-Reckoning and Particle Filter-Indoor Positioning algorithm

SK Planet Co.,Ltd., Industry-Academic Cooperation Student Researcher

Suwon, Korea

Mar - Dec 2020

- Developed an AR rhythm game application for the Busan One Asia K-pop Festival using AR Core to provide location-based interactive experiences

Awards & Honors

[Awards]

- | | |
|--|------|
| • 3 rd Place, Chung-ang University AI and Humanities Academic Paper contest | 2023 |
| • 1 st Place, Kookmin University self-driving contest | 2021 |
| • 3 rd Place, Sungkyunkwan University AI x Bookathon contest | 2021 |
| • Volunteering Excellence Prize, NIA(National Information Society Agency) | 2020 |

[Scholarships]

- | | |
|--|------|
| • Grace Hopper Conference Scholarship | 2026 |
| • Academic Excellence Scholarship (top 12%) | 2022 |
| • Creative Scholarship (100% tuition support) | 2021 |
| • Sungkyun Software Scholarship (100% tuition support) | 2019 |
| • MegastudyEdu Scholarship (external) | 2019 |

Teaching & Mentoring

[Teaching]

Graduate Teaching Assistant

Sep 2024 - Dec 2025

- *CICS 110 Introduction to Programming (Fall 2024)*: Led two lab sessions (60 students each) with hands-on instruction and grading support.
- *CS 383 Artificial Intelligence (Spring 2025)*: Mentored students in 15 group projects and assisted with assignment design and evaluation (180 students).
- *CS 571 Data Visualization and Exploration (Summer 2025)*, *INFO 348 Data Analytics with Python (Fall 2025)*: Sole TA for CS 571 (35 students) and INFO 348 (60 students); independently managed grading, office hours, and assignment design.

[Mentoring]

Undergraduate Research Volunteer Program Mentor

May - Sep 2025

- Gerindra Adi, Vidhaan Kothari, Nish Methuku (CS Undergraduate)
- Topic: Glanceable Health Visualization for Older Adults Using Large Multimodal Models

Services

IEEE VIS Poster Committee

2026

EuroVis Reviewer

2026

Skills

Technical Proficiencies: Python, Pytorch, JavaScript (React.js, D3.js, Node.js), Flask, HTML, CSS, MongoDB

Research Methodologies:

Qualitative: User Research, Semi-Structured Interviews, Prototyping, Usability Testing, Think-Aloud Protocol, Thematic Analysis

Quantitative: Survey Design, Experimental Design, Statistical Analysis, Modeling, Visualization Analysis

Tools & Platforms: Git, GitHub, Vercel, Render, Capacitor, Docker, Figma, Zeplin